



جامعة أم القرى
UMM AL-QURA UNIVERSITY

Project 6
SW/HW interface

Section: 6

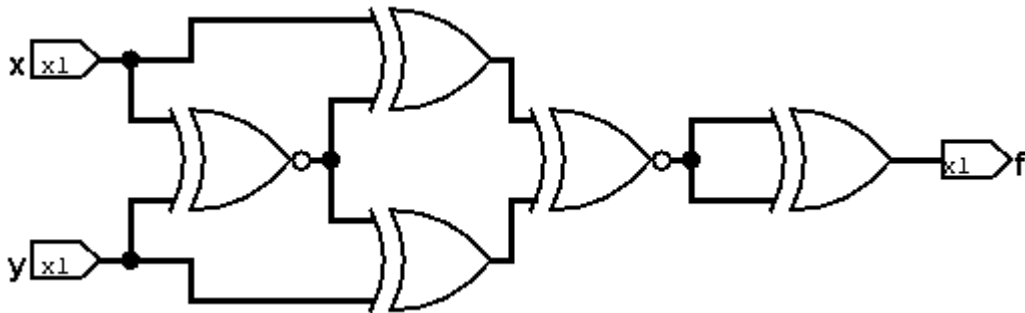
حاسوبيات 6

Team Names:

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Dana Hassan Bafaqih	445013918
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Dr. Anas Daghistani

T1. CONSTRUCT THE FOLLOWING LOGIC CIRCUIT USING [LOGISIM EVOLUTION](#).



T2. Fill in the truth table for the same logic circuit.

x	y	f
0	0	0
0	1	0
1	0	0
1	1	0

T3. Write the Boolean algebraic expression for the same logic circuit.

Let $A = x \oplus (x \oplus y)'$

Let $B = y \oplus (x \oplus y)'$

$f = (A \oplus B)' \oplus (A \oplus B)'$

T4. Minimize the circuit in T1 to the simplest possible form; then, draw the final result.

Gate Name: 0 gate

The Diagram:

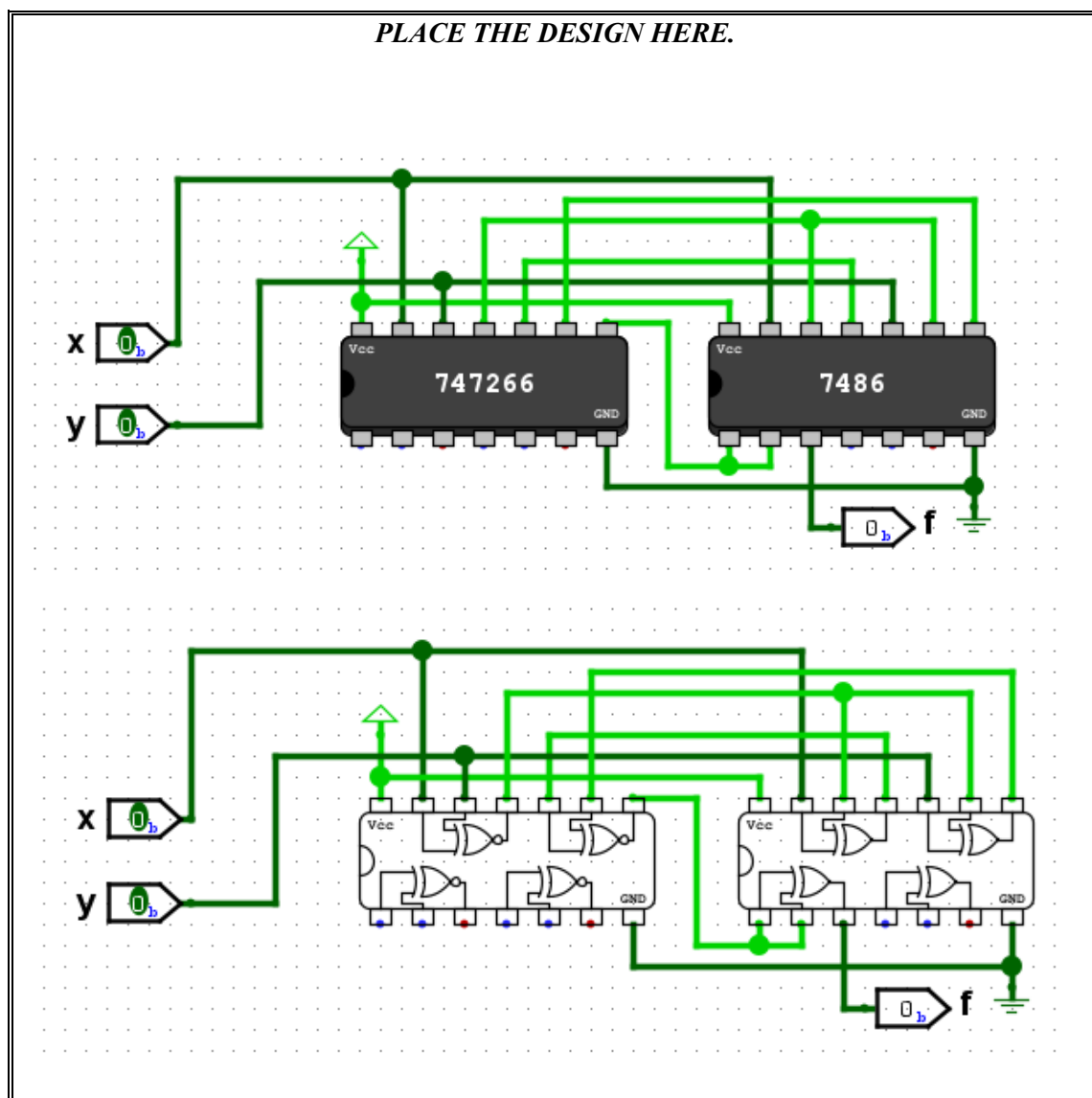




T5. BONUS: CHOOSE THE REQUIRED EQUIPMENT AND WRITE THEIR QUANTITIES TO CONSTRUCT THE DESIGN ABOVE IN T1 USING THE TTL CHIPS IN [LOGISIM EVOLUTION](#).

Quantity	Equipment
1	747266 (XNOR gate)
1	7486 (XOR gate)
2	PIN IN
1	PIN OUT, VCC, GND
7400, 7402, 7404, 7408, 7432, 7436, 7486, 747266, PIN IN, PIN OUT, VCC, GND	

PLACE THE DESIGN HERE.



Group 6 (Laptops) Comparison

الاسم الثلاثي بالعربية	دانا حسن بافقيه	بتول حسام حلاق	أمازيح حسن الزهراني	جنان حاتم بجوي	لين هاني محمدرجب
الرقم الجامعي	445013918	445013931	445007479	445000496	445001649
Test Link	Click here (Dana)	Click here (Batoul)	click here (Amareej)	click here (Jenan)	Click here (Leen)
Single-Core Score	776	1544	1223	327	1490
Multi-Core Score	1277	9322	2519	487	5415
ISA	X64	X64	X64	X64	ARMv8.6-A
User	dhbafaqih	BHhallaq	Ahalzhrani	JHbajawi	Ihamohammed
Upload Date	January 04 2025 01:08 PM	January 07 2025 02:03 PM	January 07 2025 05:08 PM	January 09 2025 12:34 PM	January 15 2025 12:18 PM
Operating System	Microsoft Windows 10 Home (64-bit)	Microsoft Windows 11 Pro (64-bit)	Microsoft Windows 11 Home (64-bit)	Microsoft Windows 11 Home (64-bit)	macOS 15.2 (Build 24C101)
Model	Dell Inc. Inspiron 5559	LENOVO 83DV	HP HP Pavilion x360 Convertible 14-dh1xxx	HP HP Laptop 15-dw1xxx	MacBook Air (2022)
Motherboard	Dell Inc. 01YVVH	LENOVO LNVNB161216	HP 866F	HP 85F5	Mac14,2
Processor Name	Intel Core i7-6500U	Intel Core i7-13650HX	Intel Core i5-10210U	Intel Celeron N4020	Apple M2
Topology	1 Processor, 2 Cores, 4 Threads	1 Processor, 14 Cores, 20 Threads	1 Processor, 4 Cores, 8 Threads	1 Processor, 2Cores	1 Processor, 8 Cores
Base Frequency	2.59 GHz	2.60 GHz	2.10GHz	1.10GHz	1.96 GHz
Max Frequency	3093 MHz	4688 MHz	4189MHz	2304MHz	1000 MHz
Codename	Skylake-U/Y	Raptor Lake	Comet Lake-U/Y	Gemini Lake	-
L1 Instruction Cache	32.0 KB x 2	32.0 KB x 10	32.0 KB x 4	32.0 KB x 2	128 KB x 1
L1 Data Cache	32.0 KB x 2	48.0 KB x 10	32.0 KB x 4	32.0 KB x 2	64.0 KB x 1
L2 (if available)	256 KB x 2	1.25 MB x 2	256 KB X 4	24 KB x 2	4.00 MB x 1
L3 (if available)	4.00 MB x 1	24.0 MB x 1	6.00 MB X 1	4.00 MB x 1	-
Memory Size	8.00 GB	16.00 GB	8.00 GB	4.00GB	8.00 GB
Frequency	1594 MT/s	4994 MT/s	2658 MT/s	2386 MT/s	6400 MT/s
Type	DDR3 SDRAM	DDR5 SDRAM	DDR4 SDRAM	DDR4 SDRAM	LPDDR5
Channels	1	2	1	1	2



1- Single-Core Benchmark(Clang)	1052 5.18 Klines/sec	1470 7.24 Klines/sec	1368 6.74 Klines/sec	571 2.81 Klines/sec	1777 8.75 Klines/sec
2- Single-Core Benchmark(Asset Compression)	604 18.7 MB/sec	1486 46.1 MB/sec	1393 43.2 MB/sec	617 19.1 MB/sec	1380 42.7 MB/sec
1- Multi-Core Benchmark(Clang)	1568 7.72 Klines/sec	16669 82.1 Klines/sec	3558 17.5 Klines/sec	912 4.49 Klines/sec	7093 34.9 Klines/sec
2- Multi-Core Benchmark(Asset Compression)	1960 60.7 MB/sec	16428 509.0 MB/sec	3865 119.8 MB/sec	857 26.5 MB/sec	5811 180.1 MB/sec

To analyze the preference between devices, we chose **programming** as the main use case. As we can see, Batoul's device (Lenovo 83DV) achieved the highest score in single-core (1544) and multi-core (9322) performance. The tests we conducted included:

- **Clang**: which measures the processor's speed in compiling code.
- **Asset Compression**: the process of reducing the size of files or data while maintaining their content.

The Lenovo 83DV achieved scores of (1486) and (16428) for single-core and multi-core, respectively, in the Asset Compression test, and (16669) in the Clang multi-core test. However, Leen's device (MacBook Air (2022)) outperformed in the single-core test with a score of (1777).

Overall, the Lenovo 83DV is considered the best device for programming among the others, as it is equipped with the Intel Core i7-13650HX processor, which has the highest speed (2.60GHz minimum and 4.68GHz maximum), representing the number of instructions the processor can execute per second. Additionally, it has the highest number of cores (14 cores), enabling the processor to execute multiple tasks or operations in parallel, and 16GB of DDR5 RAM, providing higher memory bandwidth and faster data access.